

CHAPTER FIVE

ANALYSIS OF VARIANCE

5.1. Comparison of the Mean of More than Two Populations

Analysis of variance (ANOVA) is a procedure to test the hypothesis that several populations have the same mean; i.e., it is used to test the equality of several means. The name ANOVA stems from the somewhat surprising fact that a set of computations of several variances is used to test the equality of several means.

When testing for differences in means of more than two populations, we usually do not proceed by considering all combinations of two populations at a time and testing for differences in each pair.

1. Such an approach would require several tests rather than just one.
2. If each individual test were conducted using a level of significance of say $\alpha = 0.05$, then the overall level of significance would be higher than 0.05.

Thus, we want to test simultaneously for differences among the means of all the populations, and we want the joint level of significance of the test to be α . To perform this test we make use of the F-distribution and use a method called ANOVA.

In order to use ANOVA, we assume the following:

1. All the samples were randomly selected and are independent of one another.
2. The populations from which the samples were drawn are normally distributed. If however, the sample sizes are large enough, we do not need the assumption of normality.
3. All the population variances are equal.

ANOVA is based on a comparison of two different estimates of the variances, σ^2 , of overall population.

1. The variance obtained by calculating the variation within the samples themselves – Mean Square Within (MSW).
2. The variance obtained by calculating the variation among sample means – Mean Square Between (MSB).

Since both are estimates of σ^2 , they should be approximately equal in value when the null hypothesis is true. If the null hypothesis is not true, these two estimates will differ considerably. The three steps in ANOVA, then, are:

1. Determine one estimate of the population variance from the variation among sample means
2. Determine a second estimate of the population variance from the variation within the samples
3. Compare these two estimates. If they are approximately equal in value, accept the null hypothesis.

Calculating the Variance among the Sample Means – MSB

The variance among the sample means is called Between Column Variance or Mean Square Between (MSB).

$$\text{Sample variance} = S^2 = \frac{\sum (X - \bar{X})^2}{n-1}.$$

Now, because we are working with sample means and the grand mean, let's substitute $\bar{X}$ for X , $\bar{\bar{X}}$ for $\bar{X}$, and K (number of samples) for n to get the formula for the variance among the sample means:

$$\text{Variance among sample means} = S_{\bar{X}}^2 = \frac{\sum (\bar{X} - \bar{\bar{X}})^2}{K-1}.$$

In sampling distribution of the mean we have calculated the standard error of the mean as

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}. \text{ Cross multiplying the terms } \sigma = \sigma_{\bar{X}} \sqrt{n}. \text{ Squaring both sides } \sigma^2 = \sigma_{\bar{X}}^2 n.$$

In ANOVA, we do not have all the information needed to use the above equation to find σ^2 . Specifically, we do not know $\sigma_{\bar{X}}^2$. We could, however, calculate the variance among the

sample means, $S_{\bar{X}}^2$, using $S_{\bar{X}}^2 = \frac{\sum (\bar{X} - \bar{\bar{X}})^2}{K-1}$. So, why not substitute $S_{\bar{X}}^2$ for $\sigma_{\bar{X}}^2$ and calculate an estimate of the population variance? This will give us:

$$\bar{\sigma}_{\bar{X}}^2 = S_{\bar{X}}^2 * n = \frac{\sum n(\bar{X} - \bar{\bar{X}})^2}{K-1} = \frac{n \sum (\bar{X} - \bar{\bar{X}})^2}{K-1} \text{ ---- If } n_1, n_2, \dots, n_k \text{ are equal.}$$

Which sample size to use?

There is a slight difficulty in using this equation as it stands. n represents the sample size, but which sample size should we use when different samples have different sizes? We solve this

problem by multiplying $(\bar{X}_j - \bar{\bar{X}})^2$ by its own appropriate n_j , and hence $\bar{\sigma}_{\bar{X}}^2$ becomes:

$$\text{MSB} = \bar{\sigma}^2 = \frac{\sum n_j (\bar{X}_j - \bar{\bar{X}})^2}{K-1}.$$

Where:

$\bar{\sigma}^2$ = First estimate of the population variance based on the variation among sample means (the Between Column Variance – MSB)

n_j = the size of the j^{th} sample

$\bar{X}_j$ = the sample mean of the j^{th} sample

$\bar{\bar{X}}$ = the grand mean

K = the number of samples

$K-1$ = the degrees of freedom associated with SSB.

Calculating the Variance Within the Samples (MSW)

It is based on the variation of the sample observations within each sample. It is called the within column variance or Mean Square Within (MSW). We calculate the sample

$$S^2 = \frac{\sum (X - \bar{X})^2}{n - 1}$$

variance for each sample as

Since we have assumed that the variances of the populations from which samples have been drawn are equal, we could use any one of the sample variances as the second estimate of the population variance. Statistically, we can get a better estimate of the population variance by using a weighted average of all sample variances. The general formula for this second estimate of σ^2 is:

$$\text{MSW} = \bar{\sigma}^2 = \frac{\sum_{j=1}^k (n_j - 1) S_j^2}{n_T - k}$$

Where:

$\bar{\sigma}^2$ = Second estimate of the population variance based on the variation within the samples (the Within Column Variance – MSW)

n_j = the size of the j^{th} sample

$n_j - 1$ = degree of freedom in each sample

$n_T - k$ = degrees of freedom associated with SSW

S_j^2 = The sample variance of j^{th} sample

K = the number of samples

$n_T = \sum n_j$ = the total sample size = $n_1 + n_2 + \dots + n_k$

The estimate of population variance based on variation that exists between sample means (MSB) is somewhat suspect because it is based on the notion that all the populations have the same mean. That is, the estimate MSB is a good estimate of the σ^2 only if H_0 is true and all the populations' means are equal: $\mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$.

If the unknown population means are not equal, and probably are radically different from one another, then the sample means ($\bar{X}_j$) will most likely be radically different from each other too. This difference will have a marked effect on MSB. That is to say, the $\bar{X}_j$ values will

vary a great deal and the $(\bar{X}_j - \bar{\bar{X}})^2$ terms will be large. Thus, if the population means are not all equal, then the MSB estimate will be large relative to the MSW estimate. That is, if the MSB is large relative to the MSW, and then the hypothesis that all the population means are equal is not likely to be true.

The important question is, of course, How large is “large?” also, how do we measure the relative sizes of the two variance estimates? The answer to these questions is given by the F-distribution.

If k samples of n_j ($j = 1, 2, \dots, k$) items of each are taken from k normal populations that have equal variances and for which the hypothesis $H_0: \mu_1 = \mu_2 = \dots = \mu_k$ is true, then the ratio of the MSB to the MSW is an F-value that follows an F-probability distribution.

THE F-DISTRIBUTION

$$F = \frac{MSB}{MSW}$$

Characteristics of F-distribution

1. It is a continuous probability distribution
2. It is uni-modal
3. It has two parameters; pair of degrees of freedom, v_1 and v_2
 v_1 = the number of degrees of freedom in the numerator of F-ratio; $v_1 = k - 1$
 v_2 = the number of degrees of freedom in the denominator of F-ratio; $v_2 = n_T - k$
4. It is a positively skewed distribution, and tends to get more symmetrical as the degrees of freedom in the numerator and denominator increase.

5. The mean for an F-distribution is $\frac{v_2}{v_2 - 2}$, for $v_2 > 2$; and the standard deviation is $\sqrt{\frac{2 v_2^2 (v_1 + v_2 - 2)}{v_1 (v_2 - 2)^2 (v_2 - 4)}}$ for $v_2 > 4$.

Illustration 1: The training director of a company is trying to evaluate three different methods of training new employees. The first method assigns each to an experienced employee for individual help in the factory. The second method puts all new employees in a training room separate from the factory, and the third method uses training films and programmed learning materials. The training director chooses 18 new employees assigned at random to the three training methods and records their daily production after they complete the programs. Below are productivity measures for individuals trained by each method.

Method 1	Method 2	Method 3	
45	59	41	
40	43	37	
50	47	43	
39	51	40	
53	39	52	
44	49	37	
271	288	250	
$\bar{X}_1 = 45.17$	$\bar{X}_2 = 48.00$	$\bar{X}_3 = 41.67$	$\bar{\bar{X}} = 44.94$
$S_1^2 = 30.17$	$S_2^2 = 47.60$	$S_3^2 = 31.07$	

At the 0.05 level of significance, do the three training methods lead to different levels of productivity?

Solution

1. $H_0: \mu_1 = \mu_2 = \mu_3$
 $H_1: \mu_1 = \mu_2 = \mu_3$ are not all equal

2. $\alpha = 0.05$

$v_1 = K - 1$	$v_2 = n_{T(3 \times 6)} - k$	$F_{0.05, 2, 15} = 3.68$
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$$= 3 - 1 = 2$$

$$= 18 - 3 = 15$$

Reject H_0 if sample $F > 3.68$

3. Sample F

$$MSB = \frac{\sum n_j (\bar{X}_j - \bar{\bar{X}})^2}{K - 1} = \frac{6[(45.17 - 44.94)^2 + (48.00 - 44.94)^2 + (41.67 - 44.94)^2]}{3 - 1}$$

$$= \frac{120.66}{2} = 60.33$$

$$MSW = \frac{\sum (n_j - 1) S_j^2}{n_T - K} = \frac{5[30.17 + 47.60 + 31.07]}{15} = \frac{108.84}{15} = 7.256$$

$$F = \frac{MSB}{MSW} = \frac{60.33}{7.256} = 8.31$$

4. Accept H_0 .

There is no difference in the effects of the three training programs (methods) on employee productivity.

Illustration 2: A department store chain is considering building a new store at one of the four different sites. One of the important factors in the decision is the annual household income of the residents of the four areas. Suppose that, in a preliminary study, various residents in each area are asked what their annual household incomes are. The results are shown in the accompanying table below. Is there sufficient evidence to conclude that differences exist in the average annual household incomes among the four communities? Use $\alpha = 0.01$.

Area 1	Area 2	Area 3	Area 4	
25	32	27	18	
27	35	32	23	
21	30	48	29	
17	46	25	26	
29	32	20	42	
30	22	12		
	19	18		
	51			
	27			
159	294	182	138	
$\bar{X}_1 = 26.50$	$\bar{X}_2 = 32.67$	$\bar{X}_3 = 26.00$	$\bar{X}_4 = 27.60$	$\bar{\bar{X}} = 28.63$
$S_1^2 = 26.30$	$S_2^2 = 107.5$	$S_3^2 = 136.33$	$S_4^2 = 81.30$	

Solution

1. $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$

$H_a: \mu_1, \mu_2, \mu_3, \mu_4$ are not all equal

2. $\alpha = 0.01$

$$v_1 = K - 1$$

$$v_2 = n_T - k$$

$$F_{0.01, 3, 23} = 4.76$$

$$= 4 - 1 = 3$$

$$= 27 - 4 = 23$$

Reject H_0 if sample $F > 4.76$

3. Sample F

$$MSB = \frac{\sum n_j (\bar{X}_j - \bar{\bar{X}})^2}{K - 1} = \frac{6(26.5 - 28.63)^2 + 9(32.67 - 28.63)^2 + 7(26.00 - 28.63)^2 + 5(27.60 - 28.63)^2}{4 - 1}$$

$$= \frac{227.84}{3} = 75.95$$

$$MSW = \frac{\sum (n_j - 1) S_j^2}{n_T - K} = \frac{5(26.3) + 8(107.5) + 6(136.33) + 4(81.3)}{27 - 4} = \frac{2134.68}{23} = 92.81$$

$$F = \frac{MSB}{MSW} = \frac{75.95}{92.81} = 0.82$$

4. Accept H_0 .

There is no difference exists in the average annual household incomes among the four communities.